

## Chap 7.2 (a)

$$\langle c \rangle \approx \frac{\sum_i c_i f(c_i) \Delta c}{\sum_i f(c_i) \Delta c}$$

$$\langle c \rangle = \sqrt{\frac{8k_B T}{\pi m}}$$

$$\sigma^2 \approx \sum_i (c_i - \mu)^2 f(c_i) \Delta c$$

```
In [1]: import numpy as np
import math
import matplotlib.pyplot as plt

k_B = 1.3806e-23 # in J/K
T = 298          # in Kelvin
amu_to_kg = 1.6605e-27
m = 4 * amu_to_kg # Mass of helium atom in kg

# Maxwell-Boltzmann speed distribution function
def f_MB(c, m, T):
    factor = 4 * np.pi * c**2 * (m / (2 * np.pi * k_B * T))**(3/2)
    exponent = np.exp(-m * c**2 / (2 * k_B * T))
    return factor * exponent

c = np.linspace(0, 3000, 1000)
f_c = f_MB(c, m, T)
dc=c[1]-c[0]
Area=np.sum(f_c*dc)
Mean=np.sum(c*f_c*dc)/Area
Var=np.sum(dc*f_c*(c-Mean)**2)
SD=np.sqrt(Var)
print('Mean Speed = ' +str(Mean)+ ' m/s'+ '\nSD = ' + str(SD)+ ' m/s')
```

Mean Speed =1251.5221338902634 m/s

SD = 521.7790891981626 m/s

## Chap 7.2 (b)

```
In [2]: import numpy as np
import math
import matplotlib.pyplot as plt

k_B = 1.3806e-23 # in J/K
T = [300, 350, 400, 450, 500, 550, 600] # in Kelvin
amu_to_kg = 1.6605e-27
m = 4 * amu_to_kg # Mass of helium atom in kg

# Maxwell-Boltzmann speed distribution function
def f_MB(c, m, T):
```

```

factor = 4 * np.pi * c**2 * (m / (2 * np.pi * k_B * T))**(3/2)
exponent = np.exp(-m * c**2 / (2 * k_B * T))
return factor * exponent

T_array = np.array(T).reshape(-1, 1)

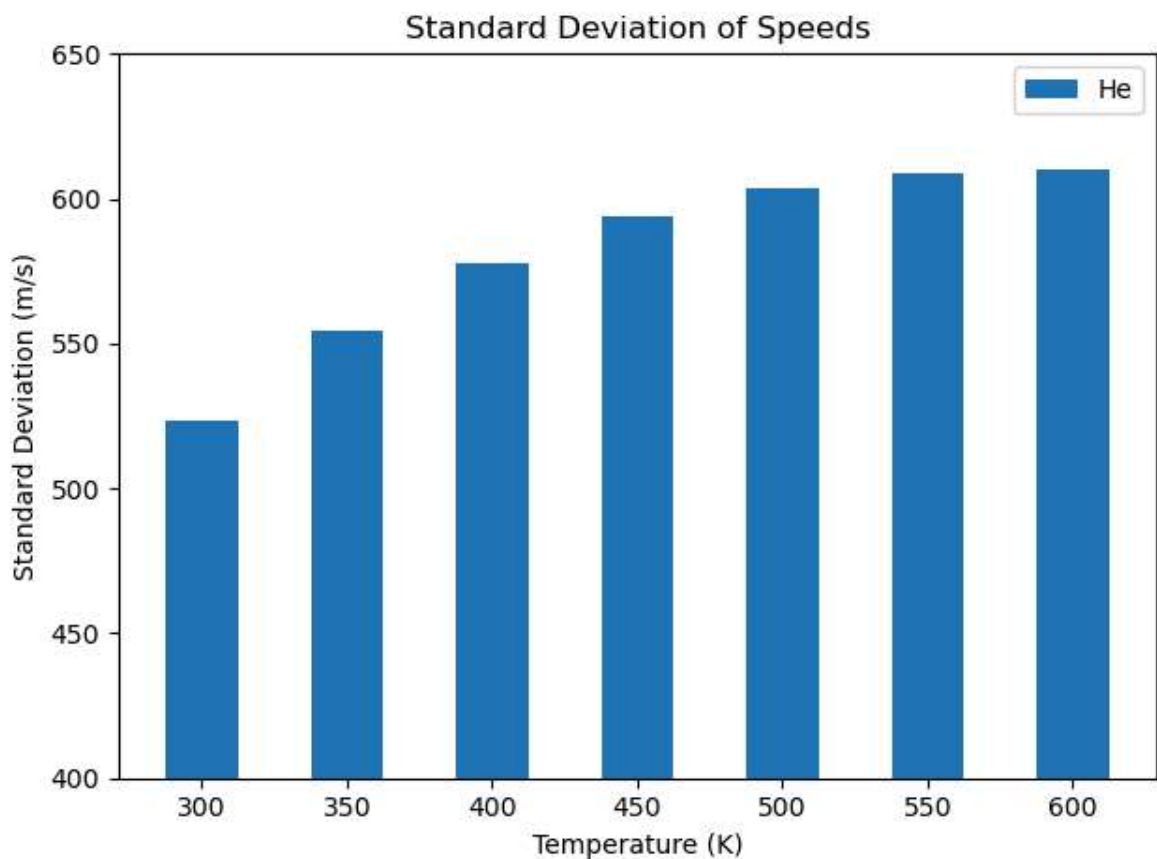
c = np.linspace(0, 3000, 1000)
f_c = f_MB(c, m, T_array)

dc=c[1]-c[0]
Area=np.sum(f_c*dc, axis=1)
Mean=np.sum(c*f_c*dc, axis=1)/Area
Var=np.sum(dc * f_c * (c - Mean[:, None])**2, axis=1)
SD=np.sqrt(Var)

plt.bar(T, SD, label='He', width=25)

plt.xlabel('Temperature (K)')
plt.ylabel('Standard Deviation (m/s)')
plt.title('Standard Deviation of Speeds')
plt.legend()
plt.ylim(400, 650)
plt.tight_layout()
plt.show()

```



## Excerise 7.3

Expectation value:  $E[X] = \frac{n+1}{2}$

Hence,  $Var[X] = E\left[\left(X - \frac{n+1}{2}\right)^2\right] = \frac{1}{n} \sum_{i=1}^n \left[\left(x_i - \frac{n+1}{2}\right)^2\right]$

Or,  $Var[X] = \frac{1}{n} \sum_{i=1}^n \left[ \left(x_i^2 - x_i(n+1) + \left(\frac{n+1}{2}\right)^2\right) \right]$

[Since,  $\sum_{i=1}^n i^2 = 1^2 + 2^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1)$

and  $\sum_{i=1}^n i = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$ ]

$Var[X] = \frac{1}{6}(n+1)(2n+1) - \frac{n(n+1)^2}{2} + n\left(\frac{n+1}{2}\right)^2 = \frac{n^2-1}{12}$

## Exerise 7.5(a)

7.5  
①

$$f(x) = \left(\frac{2}{L}\right)^{1/2} \sin\left(\frac{\pi x}{L}\right)$$

$$p(x) = \left(\frac{2}{L}\right) \sin^2\left(\frac{\pi x}{L}\right)$$

Expectation value

$$E[X] = \int_0^L x p(x) dx$$

$$= \frac{2}{L} \int_0^L x \sin^2\left(\frac{\pi x}{L}\right) dx$$

$$= \frac{2}{L} \int_0^L \left[ x \cdot \frac{1}{2} (1 - \cos\left(\frac{2\pi x}{L}\right)) \right] dx$$

$$= \frac{2}{L} \int_0^L \frac{x}{2} - \frac{2}{L} \int_0^L \frac{x}{2} \cos\left(\frac{2\pi x}{L}\right) dx$$

$$= \frac{L}{2} - 0 \quad \therefore \quad \frac{L}{2} \quad \underline{\text{Ans.}}$$

## Excerise 7.5(b)

$$\begin{aligned}
 \text{b) } \text{Var}(x) &= \int_0^L (x - \mu)^2 \frac{2}{L} \sin^2\left(\frac{\pi x}{L}\right) dx \\
 &\Rightarrow \int_0^L \left(x - \frac{L}{2}\right)^2 \frac{2}{L} \cdot \frac{1}{2} (1 - \cos\left(\frac{2\pi x}{L}\right)) dx \\
 &\Rightarrow \frac{1}{L} \int_0^L \left(x - \frac{L}{2}\right)^2 (1 - \cos\left(\frac{2\pi x}{L}\right)) dx \\
 &\Rightarrow \frac{1}{L} \int_0^L \left(x - \frac{L}{2}\right)^2 dx - \frac{1}{L} \int_0^L \left(x - \frac{L}{2}\right)^2 \cos\left(\frac{2\pi x}{L}\right) dx \\
 &\Rightarrow \frac{L^2}{12} - \frac{L^2}{2\pi^2} \\
 \therefore \text{Standard deviation or Uncertainty} &= \sqrt{\text{Var}(x)} \\
 &= \frac{L}{\sqrt{12} - \frac{1}{2\pi^2}} \quad \text{Ans.}
 \end{aligned}$$

## Excerise 8.4

```

In [4]: import numpy as np
X=[-2,-1,0,1,2]
Y=[5,2,4, 1,3]
covariance=np.cov(X,Y)
print("Covariance matrix\n", covariance)
pearson=np.corrcoef(X, Y)
print("\nPearson correlation coefficient=", np.round(pearson[0,1], decimals=1))

```

Covariance matrix

```

[[ 2.5 -1.25]
 [-1.25 2.5 ]]

```

Pearson correlation coefficient= -0.5

In [ ]:

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